ICC

SAP SE

Test Plan

S/4-HA-CLU 1.0

SAP Application Server HA Interface Certification

SAP S/4HANA 1809

Version 1.06

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1 About the SAP Application Server HA Interface Certification

The SAP Application Server HA-Interface-Certification for HA Partners certifies the Start/Stop infrastructure within HA-Setups (SAP NetWeaver Management Agents, Cluster-API).

The overall goal of the HA-Interface-Certification is to establish a SAP HA-Setup scenario that will be known to all HA partners and will be the reference. More specifically for this certification, it is the intention to unify HA-Setups for ABAP platform 1809, which is the underlying technology platform for SAP S/4HANA 1809. In this setup the Standalone Enqueue Server 2 together with the Enqueue Replicator 2 is used. They are available since SAP NetWeaver 7.52 for ABAP.

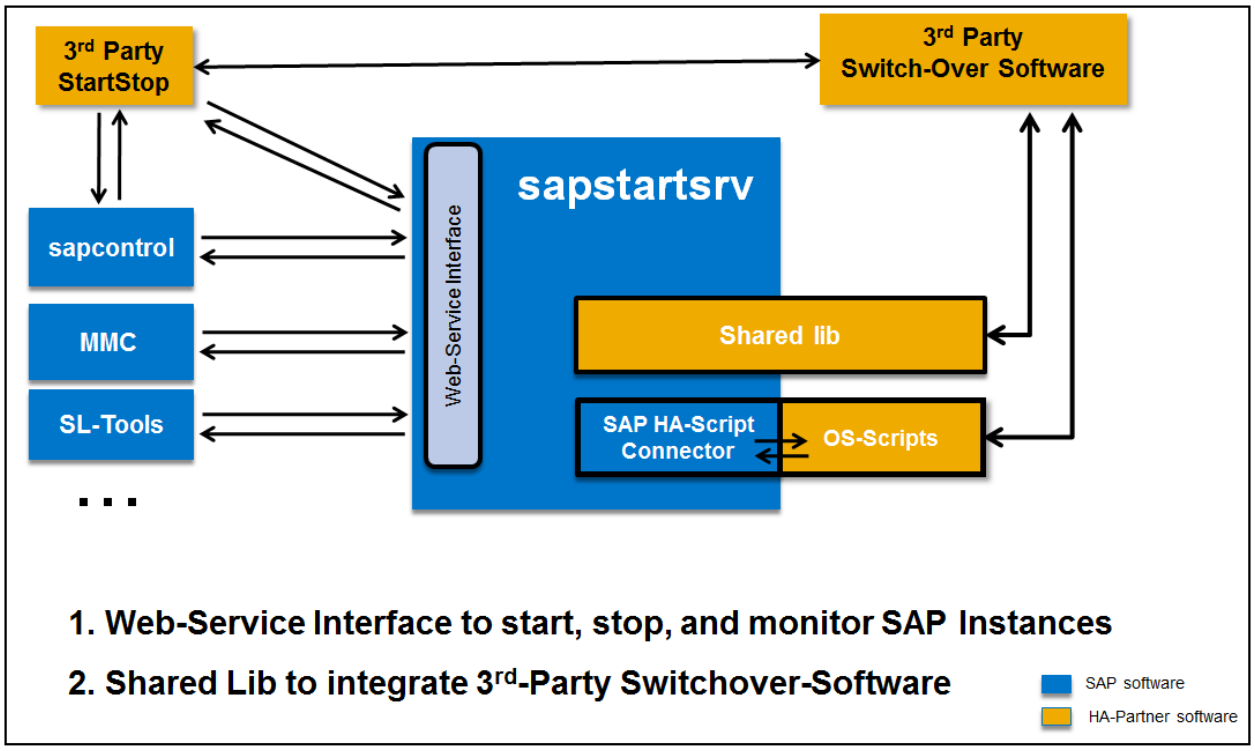
To test the old enqueue mechanism there is a second HA certification named NW-HA-CLU 750 available that uses an HA-Setup for SAP NetWeaver 7.5 for ABAP as well as for Java based application servers. But this certification is not subject of this document.

It is expected to improve the transparency about operating system dependencies and 3rd party failover software as well as to improve the transparency about available SAP HANA-specific HA-Solutions of 3rd party vendors. Additionally, the responsibilities of SAP and its partners, especially for support issues will be clarified.

Important Resources for partners about the HA-Interface Certification can be found here.

1.1 Objectives of this certification

The following SAP Start Service (sapstartsrv) interface configuration provides functionality that the HA vendor solution must be able to use in a customer environment:



The following tests will help to verify the integration of the 3rd party software in different situations.

1.2 Certification criteria

The following certification criteria will verify the correct functionality and general readiness of the third-party HA software operating the various SAP Software components via the HA-Interface offered by the SAP Start Service (sapstartsrv):

- **General Certification Criteria**
  1. Partner commits to regularly contact SAP for changes in the SAP recommendations for SAP HA-Setup, at the latest before stating support for a new major SAP NetWeaver or SAP S/4HANA release.
  2. SAP-specific documentation is available at partner web site for targeted user groups (at least to basis consultants and customers, if possible, publicly accessible).
  3. Support responsibilities for HA-Setups are defined and agreed in writing within a dedicated SAP note and are released to customers.
  4. Partner informs whether an HA-Setup is offered for SAP HANA database.
- **Technical Certification Criteria**
  1. Name and version of 3rd-party HA software is delivered correctly.
  2. HA configuration is verified successfully.
  3. Successful failover can be performed manually via 3rd-party software.
  4. Crucial SAP-processes are monitored and restored automatically (Restart and/or failover).
  5. Usage of sapstartsrv and integration into 3rd-party failover solution
  6. sapstartsrv status information is integrated into 3rd-party failover detection.
  7. Starting an instance must be possible via standard web service interface (SAPControl).
  8. Stopping an instance must be possible via standard web service interface (SAPControl) without triggering an instance failover.
  9. A document is maintained describing on how to temporarily disable failover software in case of SW maintenance activities. If this is not possible or very difficult, it should describe on how to perform maintenance activities in a fail-over environment. All software maintenance activities that are not possible should be documented as well.

2 The certification procedure in detail

2.1 Preparing the environment

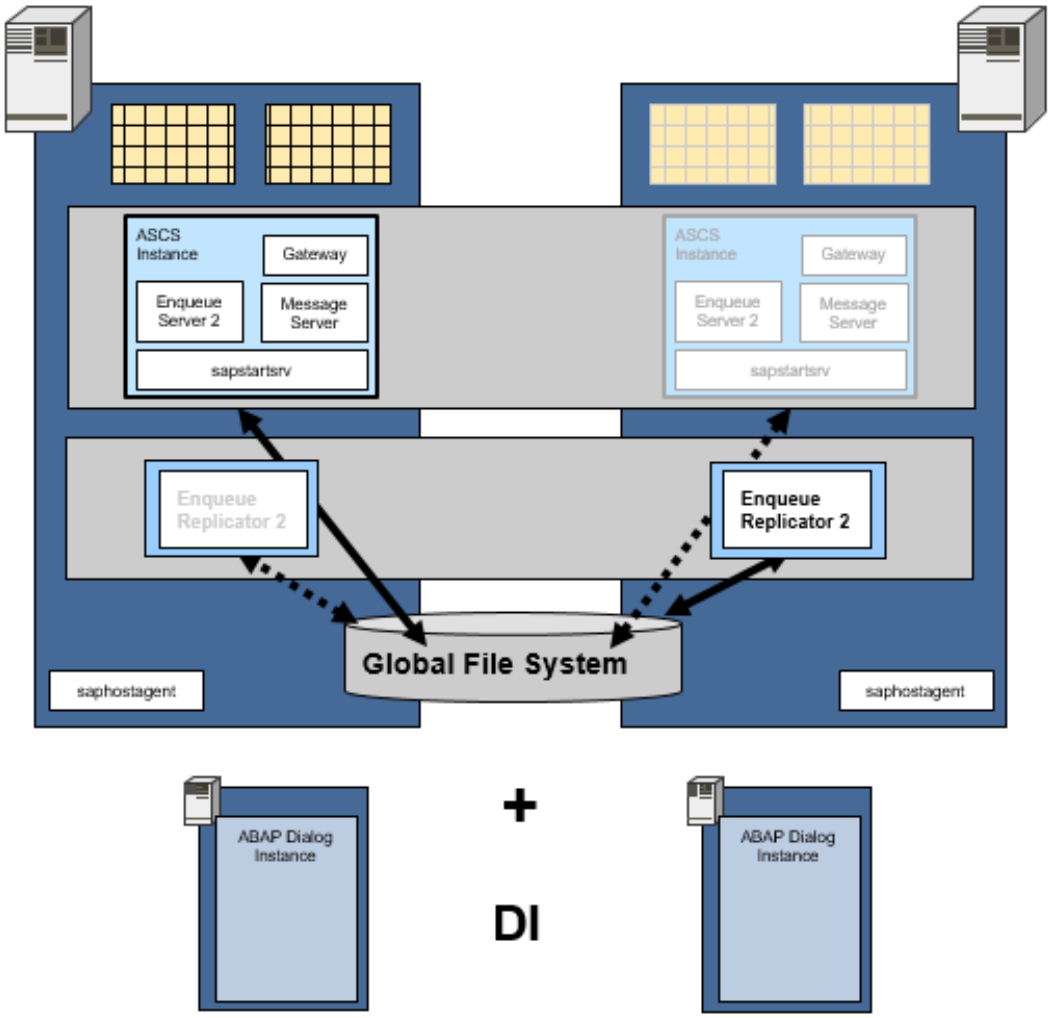
2.1.1 Installing the HA-Solution

The HA-Partner needs to complete all steps required to install the HA-Solution on respective hosts. The HA-Solution must be fully functional and accessible for the certification process.

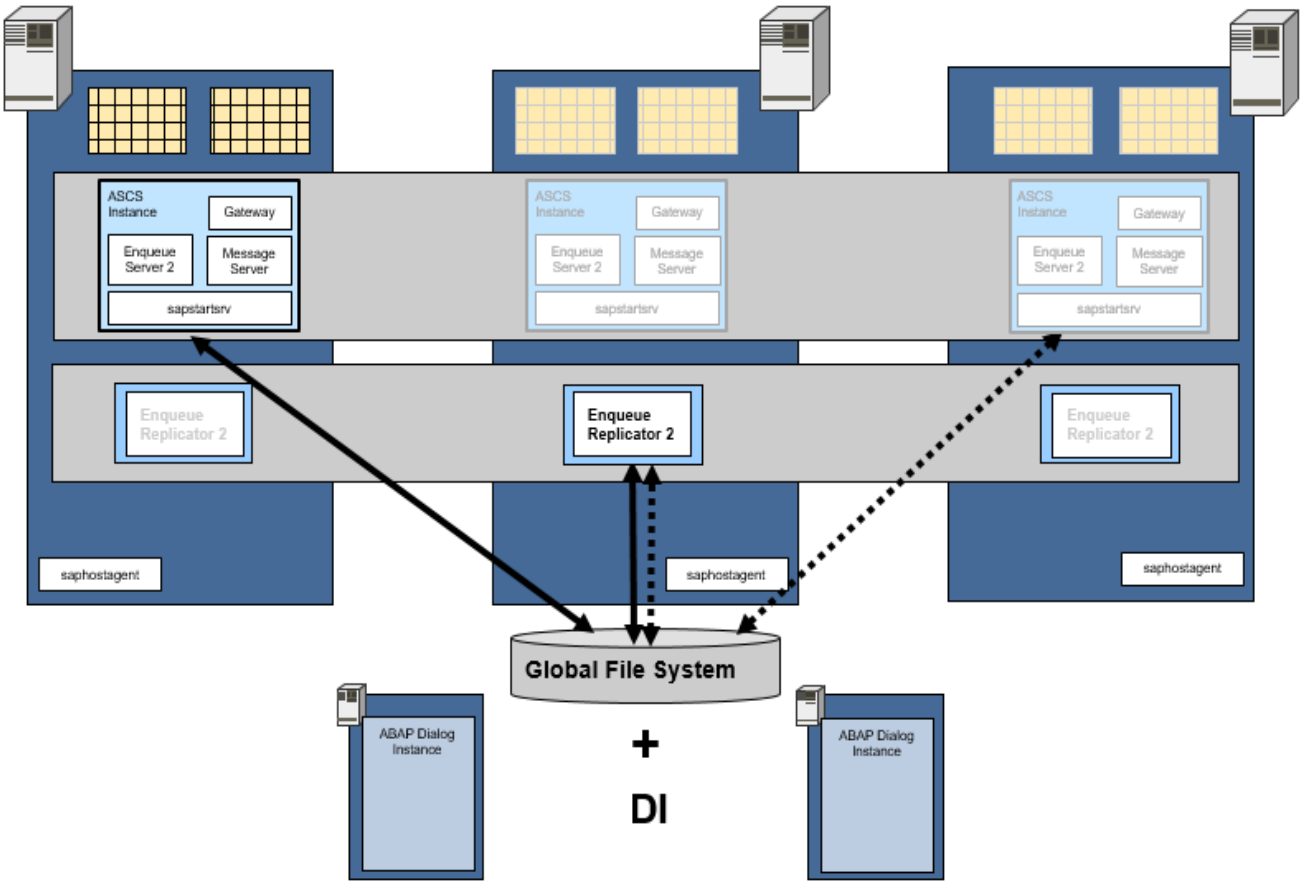
2.1.2 Installing the SAP-System

The recommended behavior in case of failures differs dependent on the number of cluster nodes used for the Standalone Enqueue Server 2 together with the Enqueue Replicator 2. Therefore, two different HA-Setups are the basis of this HA certification: One with 2 cluster nodes and one with more than two cluster nodes. It is necessary to setup both environments with the essential components for the HA certification.

2-Nodes Cluster



n-Nodes Cluster (n > 2)



Before the actual certification process starts, the test environment must be set up by the HA-Partner as follows:

1. An ABAP platform 1809 system must be installed for the certification process. As minimum kernel version SAP Kernel 773 PL 22 + ENSA2_29-xxxxxx.SAR is required. SAP Kernel 773 PL100 and above already contain the ENSA2 patch. Because one of the technical tests requires the exchange of the kernel by a newer one, make sure that you don't start with the newest available kernel patch level.

   The system must be set up by the partner for high availability according to the partner's documentation. A 2-nodes cluster and a cluster with more than two nodes must be provided for executing the certification tests. The test environment can be simulated using virtualization technology.

2. The following SAP components and instances must be installed and setup correctly:
   a. A separated ASCS instance, manageable in a dedicated switch-over group
   b. The Enqueue Replicator 2 (ER2) is setup and manageable in a dedicated switch-over group
   c. The hosts for the ASCS and the ER2 switch-over group should be the same
   d. At least 2 ABAP-Dialog instances are installed on two (virtual or physical) hosts, the hosts of the dialog instances can be used also to hold the ASCS and/or ERS-setup

   e. sapstartsrv is part of the switch-over group and must be moved as well in case of a failover
   f. saphostagent installed once per host and not part of the failover group
   g. Platform dependent SAPCPE for replication of the executables is in place

3. Independently of whether the SAP System is to be integrated into the HA-Solution by using the HA-Interface-Script-Connector or a vendor HA library, the required SAP Kernel patch level must be applied as described in SAP note 1693245.
4. The HA vendor software is up and running and controls the SAP System as desired.
5. An SAP Management Console (SAP MMC or SAP MC) must be installed to monitor the system landscape. All instances must be visible there and should be running properly. To ease the operational tasks, the SAP MMC/MC should be called from a separate machine (frontend machine) to be independent from switching the cluster nodes.

In addition to these mandatorily required setup-characteristics for the HA-Interface certification, the hardware as well as the database is typically set up in a highly available manner (vendor specific) at customer side. To pass the certification, it is not necessary to setup the hardware or database in a highly available manner though, because these components are not in the scope of HA-Interface certification at all.

2.1.3 Installing the HA Test Tool

For some of the tests you need the SAP HA Test Tool. At least version v2.14 must be installed. The installation bundle is not part of ABAP platform 1809. It can be received from SAP ICC.

sapcontrol (is part of the SAP Kernel) is needed by the HA Test Tool to run the tests. It must be accessible on the client-side of the tool. If you use the host on which a dialog instance of the SAP system is installed for the client part of the tool, you can use sapcontrol of the dialog instance. If the operating system used for the client-side of the tool differs from the operating system of the application servers of the ABAP platform 1809 installation, the appropriate SAP Kernel (see chapter 2.1.2, section 1.) can be downloaded from the SAP Service Marketplace.

2.2 General tests to certify the HA-Solution

Test	Title
GEN01	Recommended SAP HA-Setup is described in vendor specific documentation
GEN02	SAP-specific documentation is available at partner web site
GEN03	Support processes are documented and available for customers
GEN04	Inform whether SAP HANA database is supported by the HA-Solution

2.2.1 Test GEN01: Recommended SAP HA-Setup is described in vendor specific documentation

Purpose of the test:

Check whether the recommended SAP HA Setup is available and part of the documentation delivered to customers via the HA vendor documentation.

Test procedure:

1.) HA-Partner to provide an URL to this SAP specific setup documentation to SAP during certification

2.) Tester to check, whether the recommended HA-Setup by SAP is described appropriately.

Expected outcome:

- Recommended HA-Setup by SAP is described in vendor specific documentation

2.2.2 Test GEN02: SAP-specific documentation is available at partner web site

Purpose of the test:

Check whether the documentation of the HA-Partner describes all required setup and maintenance requirements for SAP products is publicly available for customers. It must be accessible for the HA vendor customers at least, and it is important that customers that acquired the HA vendor software have always the possibility to access these documents.

Test procedure:

1.) HA-Partner to provide an URL to this SAP specific documentation to SAP during certification

2.) Documentation to be accessed by the tester

Expected outcome:

- SAP-specific documentation is publicly available at partner web site

2.2.3 Test GEN03: Support processes are documented and available for SAP customers

Purpose of the test:

For certified HA-Partners, SAP requires that the support processes are defined and agreed in writing. Especially for customers who set up their SAP System in a highly available manner, it is crucial, that in case of an outage or any other related support issue, the tickets are processed effectively from both support organizations.

Therefore, SAP developed an HA-Support questionnaire, which must be filled out in agreement with both support organizations of SAP and the HA-Partner.

Test procedure:

1.) Along the registration process, the HA-Support questionnaire will be sent to the SAP Support organization to document the respective support processes with the respective HA-Partner.

2.) SAP Support and the support organization of the HA-Partner are supposed to document the support processes according to current agreements. In addition to the HA-Support questionnaire the outcome will be documented within an SAP note. The note should be structured according to the document HASupportNoteStructure v2.01.pdf and must be in status “Released to customers”.

3.) Tester to check whether the HA-Support questionnaire is filled out completely, and the corresponding SAP note is available for customers and describes the support process with the HA-Partner for the respective operating system.

Expected outcome:

- HA-Partner will provide this information in the support document that needs to be agreed from both sides (HA-Partner and SAP) before the certification process is finished.
- Respective SAP note describes the agreed support procedures according to the specified structure and is in status “Released for customers”.

2.2.4 Test GEN04: Inform whether SAP HANA database is supported by the HA-Solution

Purpose of the test:

SAP HANA database is the only database used by SAP S/4HANA 1809. Any customer, who considers implementing a recommended HA-Setup, has to decide whether the database should also be implemented in a highly available manner. It is important for the customer to know whether the same HA-Solution could be used for the SAP system as well as for the database.

Therefore, the HA-Partner for the given HA-Solution should provide the information whether SAP HANA database is supported, including the list of supported SAP HANA database versions. The operating system used for the S/4-HA-CLU certification might be different than the operating system used for SAP HANA database.

Test procedure:

1.) HA-Partner to provide the information whether SAP HANA database is supported, including the list of supported SAP HANA database versions. A link to the documentation of the HA-Solution for SAP HANA database has to be provided as well.

2.) In case SAP HANA database is supported by the HA-Solution, the tester must check, whether the provided documentation informs about the SAP HANA database versions handled by the HA-

Solution.

3.) Tester to document the currently supported database versions at the time when the certification takes place.

Expected outcome:

- Information whether the HA-Solution supports SAP HANA database, and if so, which database versions are supported. Additionally, a link to the HA documentation must be provided.

2.3 Technical tests to certify the HA-Solution

The following tests verify the functional correctness of the HA-Solution in conjunction with the HA-Interface of the SAP Start Service (sapstartsrv).

Each test can be performed several times to verify the repeatability, reliability and completeness of the solution.

If not indicated otherwise, every test can be performed manually using the Web-Service-Interface or using the functionality provided by the HA-Solution, if available.

If not stated otherwise, a test can be executed on one of the two HA setups (see chapter 2.1.2).

Test	Title	HA Setup (# of nodes)
TEC01	Name and version of 3rd-party HA software is delivered correctly	
TEC02	Verification of HA configuration shows no errors	
TEC03	Shared Lib (HA-Interface) is being loaded without any errors	
TEC04	Manually moving the ASCS works correctly	2
TEC05	Irrecoverable outage of the Enqueue Server 2 is handled correctly	2
TEC06	Outage of the Enqueue Replicator 2 is handled correctly	2
TEC07	ASCS moves correctly under load without specifying the destination node	>2
TEC08	ASCS moves correctly in case of hardware or OS failure	>2
TEC09	Recoverable outage of the Message Server is handled correctly (if the SAP Profile Parameter “Restart_Program” is used for the Message Server)	>2
TEC10	Irrecoverable outage of the Message Server is handled correctly (if the SAP Profile Parameter “Start_Program” is used for the Message Server)	>2
TEC11	Enqueue Replicator 2 moves correctly in case of hardware or OS failure	>2
TEC12	Starting and stopping of SAP resources work correctly	
TEC13	Outage / Restart of the SAP Start Service is handled correctly	
TEC14	Automated ASCS restart in place works correctly	
TEC15	Automated execution of the Rolling Kernel Switch is possible	
TEC16	Simulating an upgrade works correctly using the Maintenance Mode (if the functionality is available)	
TEC17	Simulating an upgrade works correctly (if no Maintenance Mode is available)	
TEC18	Simulating cluster failure works correctly (if the functionality is available)	

For all test cases, monitoring of the SAP System is crucial, in particular that all instances are available again after the switch-over procedure of an instance. It is of particular interest whether

- the Enqueue lock table is correctly restored (Same entries available in restored table, see appendix 3.1: How to check whether the lock table is restored correctly),
- the Enqueue Replication Service is up and running afterwards and the lock table replication is re-established again, and
- the central file systems must be available after the switch again.

2.3.1 Test TEC01: Name and version of 3rd-party HA software is delivered correctly

Purpose of the test:

The name and the version of the used HA software is important information when the configuration of an SAP system should be shown.

Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated.

Test procedure:

1.) The HA-Partner must provide name and version of the HA-Solution used for the test system.

2.) Use SAP MC/MMC to get the information about name and version of the HA-Solution.

Expected outcome:

- Name and version of the HA-Solution delivered by SAP MC/MMC has to match the information delivered by the HA-Partner.

2.3.2 Test TEC02: Verification of HA configuration shows no errors

Purpose of the test:

The HA-Interface provides checks to verify that the HA environment is set up correctly.

Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated.

Test procedure:

1.) Use SAP MMC or the HACheckConfig function of sapcontrol to verify the HA configuration of the SAP system.

Expected outcome:

The result of all checks must be ‘Success’. No error must be shown.

2.3.3 Test TEC03: Shared Lib (HA-Interface) is being loaded without any errors

Purpose of the test:

The vendor specific Shared Lib provided by the HA-Partner is supposed to be loaded correctly, also in case the SAP HA-Script-Connector is used.

Prerequisites:

- None.

Test procedure:

1.) Analyze dev trace of sapstartsrv and log of cluster software for corresponding log entries

2.) Check whether the HA-Script-Connector is configured correctly for those instances being controlled by the HA-Solution.

3.) The library is to be loaded from the local directory, which is not affected by a failover.

Expected outcome:

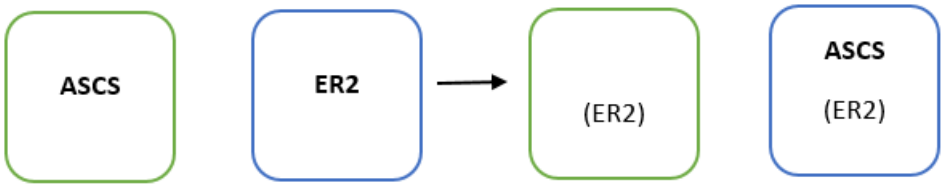
- No errors can be found in the logfile …/work/sapstartsrv.log of the active ASCS and ERS instances.

2.3.4 Test TEC04: Manually moving the ASCS works correctly

Purpose of the test:

It is the purpose of a HA-Solution to move the ASCS from one cluster node to another. A manual move of the ASCS is considered as basic operation within the cluster. This test ensures, that the manual switch can be performed without any errors.

Illustration:



(ER2) means that no statement concerning ER2 movement is done in this test. ER2 could stay on the second node or move to the first one.

Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated. The HA Setup with 2 cluster nodes must be used.

Test procedure:

1.) Fill the Enqueue-Locktable as described in chapter 3.1.

2.) Tester to trigger a fail-over via the user interface of the HA-Solution

Expected outcome:

- The ASCS moves correctly to the second cluster node and the Enqueue-Locktable is correctly taken over by the ENQ-Server (see appendix 3.1: How to check whether the lock table is restored correctly).

2.3.5 Test TEC05: Irrecoverable outage of the Enqueue Server 2 is handled correctly

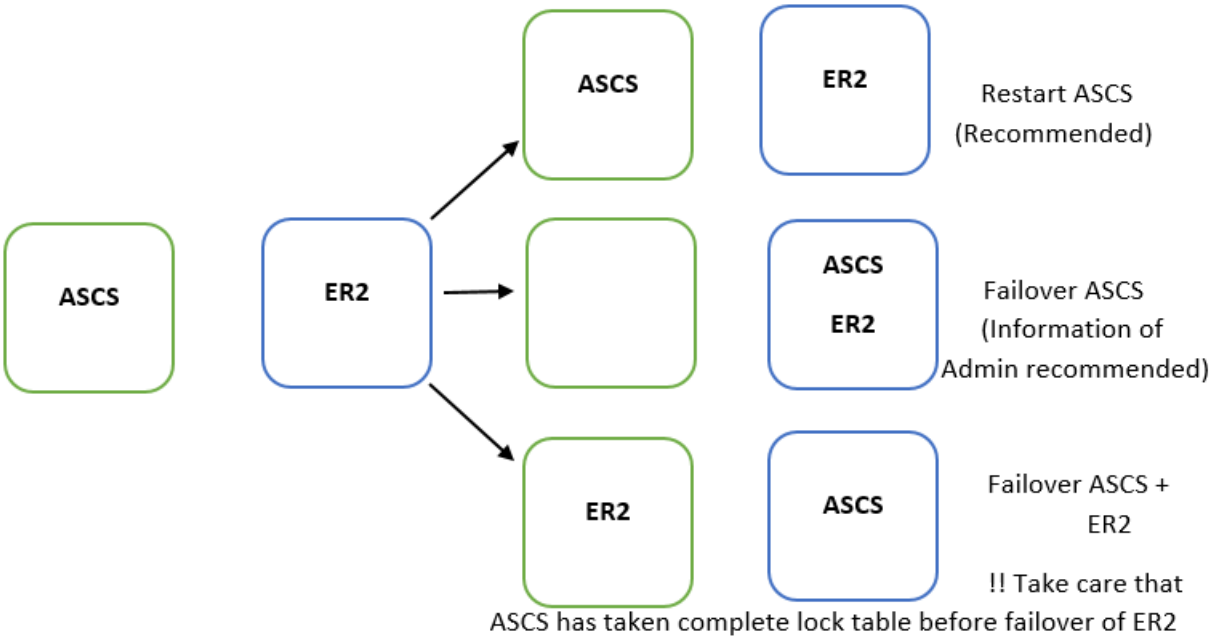
Purpose of the test:

The Enqueue Server 2 is a Single Point of Failure (SPOF) within the SAP System. Therefore, it is crucial to secure the Enqueue Server 2 within a switchover group and ensure that the Enqueue lock table is taken over by the restarted Enqueue Server 2 process correctly.

Purpose of this test is it, to verify, that in case of a failure of the Enqueue Server 2, the HA-Solution detects the failure properly, and triggers a switchover or restarts the ASCS instance with the Enqueue Server 2 process correctly.

As described in SAP note 768727, the SAP Startup Framework does NOT restart the Enqueue Server 2 process automatically.

Illustration:



Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated. The HA Setup with 2 cluster nodes must be used.

Test procedure:

1.) Verify that the SAP system runs in a stable mode, and that HA-Solution is activated.

2.) Note down the process ID of the Enqueue Server 2 (ES2)

3.) Fill the Enqueue-Locktable as described in chapter 3.1.

4.) Kill the Enqueue Server 2 process on operating system level

Expected outcome:

Two reactions are allowed for the HA-Solution:

1.) The HA-Solution restarts the ASCS, which takes over the lock table from the Enqueue Replicator 2 (ER2). This means that the ASCS instance remains on the same node as before. Only if the restart

fails a certain amount of times, a failover of the ASCS is triggered.
This is the recommended reaction.

2.) The HA-Solution triggers a failover of the ASCS to the cluster node, on which the ER2 runs without trying to restart it before.

   a. The ASCS is started and takes over the lock table from the ER2.

   b. Optionally, the HA-Solution starts the ER2 on the node where the Enqueue Server 2 was running before it was terminated. In that case, the HA-Solution has to take care, that the lock table was fully transferred to ES2 before. This can be checked via the tool enq_admin.
   ```
   enq_admin --replication_state pf=<ASCS Instance profile>
   ```
   For the usage see: enq_admin --help. The call will return '0' if the lock table is in sync again after delivering returncodes > 0 during failover.

   c. If ES2 stays on the same node than ER2, it is recommended that the HA-Solution informs the administrator about this situation.

Test result OK if:

- The lock table is correctly restored (see appendix 3.1: How to check whether the lock table is restored correctly).
- ER2 is available and the replication table must contain the same locks than the Enqueue-Locktable (see appendix 3.1: Double-check the locks held by the Enqueue Replicator 2 (ER2)).
- Central file systems are available.
- The Enqueue Server 2 and the Enqueue Replicator 2 must not be restarted by SAP.

Comments:

- The logfile for the Enqueue Server 2 is: dev_enq_server.
- The logfile for the Enqueue Replicator 2 is: dev_enq_replicator.

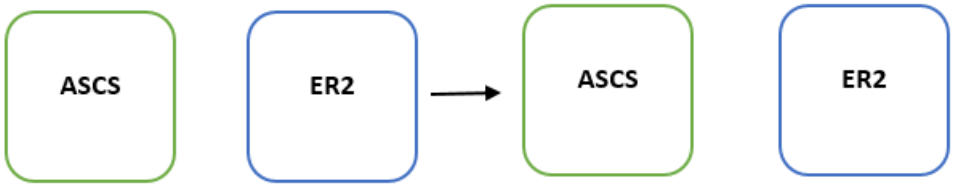
2.3.6 Test TEC06: Outage of the Enqueue Replicator 2 is handled correctly

Purpose of the test:

The Enqueue Replicator 2 plays a crucial role in the HA-Setup of the SAP System. It ensures that the lock-table is replicated to one or several different hosts and thus can be recovered by the Enqueue Server 2 in case of a failover.

The HA-Solution is supposed to monitor the availability of at least one ER2 process and should trigger an automatic restart if the ER2-Service is stopped. In that case the Enqueue Replicator 2 must not be restarted by the SAP system (parameter Restart_Program_nn not allowed in the start profile of the Enqueue Replicator 2).

Illustration:



Prerequisites:

- HA-Solution is up and running including ER2. The HA Setup with 2 cluster nodes must be used.

Test procedure:

1.) Verify that the SAP system runs in a stable mode, and that HA-Solution is activated.

2.) Note down the process ID of the Enqueue Replicator 2

3.) Fill the Enqueue-Locktable as described in chapter 3.1.

4.) Kill manually the active ER2 process on operating system level

Expected outcome:

- The ER2 service is automatically restarted by the HA-Solution. Only if the restart fails a certain amount of times, a failover of the ER2 is triggered.
- The existing enqueue table is replicated to the ER2.

Test result OK if:

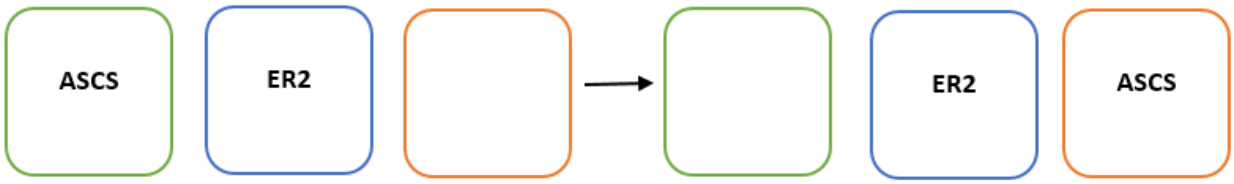
- ER2 is available and replication is re-established. Therefore, the ER2 has to be restarted automatically by the HA-Solution.
- The replication table must contain the same locks than the Enqueue-Locktable (see appendix 3.1: Double-check the locks held by the Enqueue Replicator 2 (ER2)).
- The Enqueue Replicator 2 must not be restarted by SAP

2.3.7 Test TEC07: ASCS moves correctly under load without specifying the destination node

Purpose of the test:

It is the purpose of a HA-Solution to move the ASCS from one cluster node to another. The HA interface provides the possibility to trigger a fail-over without specifying a destination node. In that case, it is up to the HA-Solution to choose the destination node and move the ASCS to it. If possible, a destination node without an active Enqueue replicator 2 should be chosen. This test ensures, that the switch can be performed without any errors. Additionally, to simulate a real-world scenario, the fail-over happens while the server system is under load.

Illustration (Example with 3 nodes):



Prerequisites:

- Verify that the HA Test Tool is installed, that the SAP system runs in a stable mode, and that the HA-Solution is activated. The HA Setup with 3 or more cluster nodes must be used. sapcontrol must be installed on the client side.

Test procedure:

1.) HA-Partner provides Tester with the configuration, event and check files for the test according to the

templates of chapter 3.2.

2.) Tester runs the HA Test Tool to trigger a fail-over via sapcontrol.

Expected outcome:

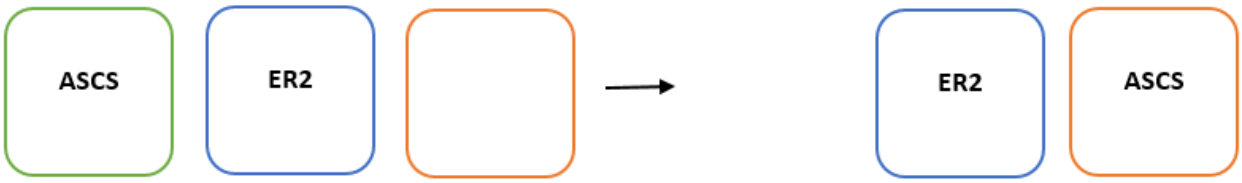
- The ASCS moves correctly to another cluster node, but not using the node with the active ER2.
- The HA Test Tool terminates with exit code 0 and just shows a period of long response times for the duration of the fail-over

2.3.8 Test TEC08: ASCS moves correctly in case of hardware or OS failure

Purpose of the test:

The HA-Solution is usually also setup to take care about hardware outages and/or failures of the operating systems. Therefore, this test checks whether an unexpected hardware or OS failure triggers a restart of an ASCS instance on a different node.

Illustration (Example with 3 nodes):



Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated. The HA Setup with 3 or more cluster nodes must be used.

Test procedure:

1.) Fill the Enqueue-Locktable as described in chapter 3.1.

2.) Turn off the power of the machine or stop the machine, on which the ASCS resides.

Expected outcome:

- The HA-Solution automatically triggers a failover of the ASCS to another cluster node, but not using the node with the active ER2.
- The move of the ASCS to the other cluster node is performed correctly.
- The lock table is correctly restored (see appendix 3.1: How to check whether the lock table is restored correctly).

2.3.9 Test TEC09: Recoverable outage of the Message Server is handled correctly (if the SAP Profile Parameter “Restart_Program” is used to start the Message Server)

Purpose of the test:

The message server is a Single Point of Failure (SPOF) within the SAP System. Therefore, it is crucial to secure the message server within a switchover group.

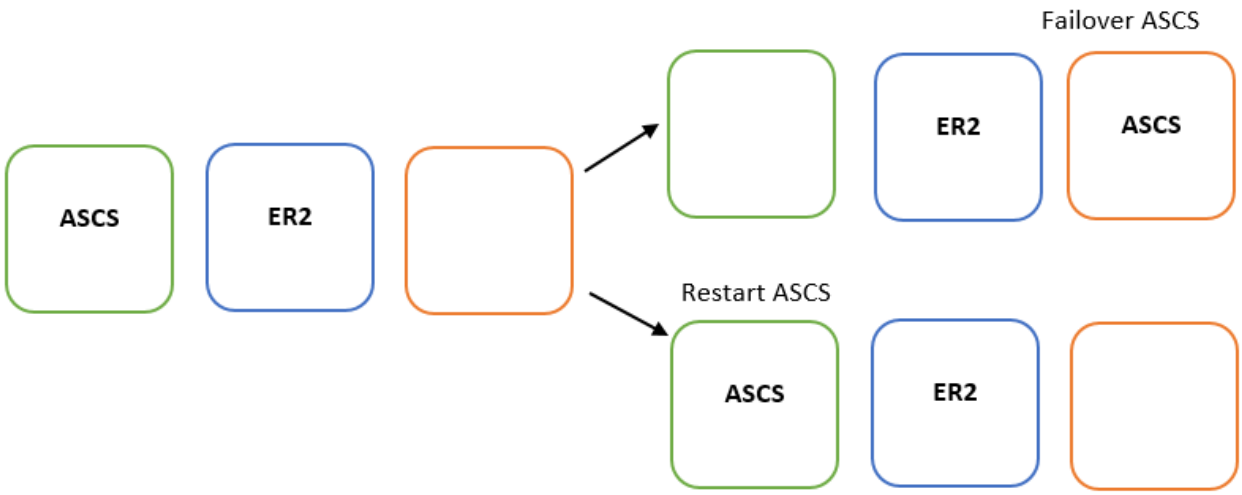
Purpose of this test is it, to verify, that in case of a failure of the message server, the HA-Solution detects

the failure properly, and triggers a switchover or restart correctly if necessary.

The SAP Start Service offers by default a mechanism to restart the message server automatically. Details can be found here.

This test also has the purpose to verify, that these capabilities work well together with the HA-Solution and do not interfere in an unwanted manner.

Illustration (Example with 3 nodes):



Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated. The HA Setup with 3 or more cluster nodes must be used.
- Check whether the SAP Profile Parameter "Restart_Program" is used to start the message server. If it is not used, this test is obsolete, and test TEC10 has be executed instead. If it is used, follow the test procedure below.

Test procedure:

1.) Kill the message server process on operating system level manually once. The message server should be restarted automatically by the SAP Start Service.

2.) Repeat killing the message server process, until it is no longer automatically restarted, depending on the configuration of the built-in restarting mechanism.

Expected outcome:

- The Message Server should be restarted automatically by the SAP Startup framework without triggering a failover (Recoverable Error, no fail-over) as often as configured in the profile parameter.
- The process ID should have changed each time, and the restart of the message server should be logged correctly in the sapstartsrv.log or sapstart.log (UNIX).

If the automatic restarting functionality does not restart the message server any longer, two reactions are allowed for the HA-Solution:

1.) The HA-Solution restarts the ASCS. This means that the ASCS instance remains on the same node as before. Only if the restart fails a certain amount of times, a failover of the ASCS is triggered. The target node must not be the node with the active ER2.

2.) The HA-Solution triggers a failover of the ASCS instance automatically. The target node must not be the node with the active ER2.

Comments:

- On Windows, the SAP profile parameter Restart_Program causes to stop restarting the process when the process dies five times within five minutes.
- On Unix, the SAP profile parameter Restart_Program typically causes to stop restarting the process when the process dies five times within ten minutes, if the Parameter <Max_Restart_Program> is not set otherwise.
- Killing the message server five times should cause an unrecoverable error of the message server.

2.3.10 Test TEC10: Irrecoverable outage of the Message Server is handled correctly (if the SAP Profile Parameter "Start_Program" is used to start the Message Server)

Purpose of the test:

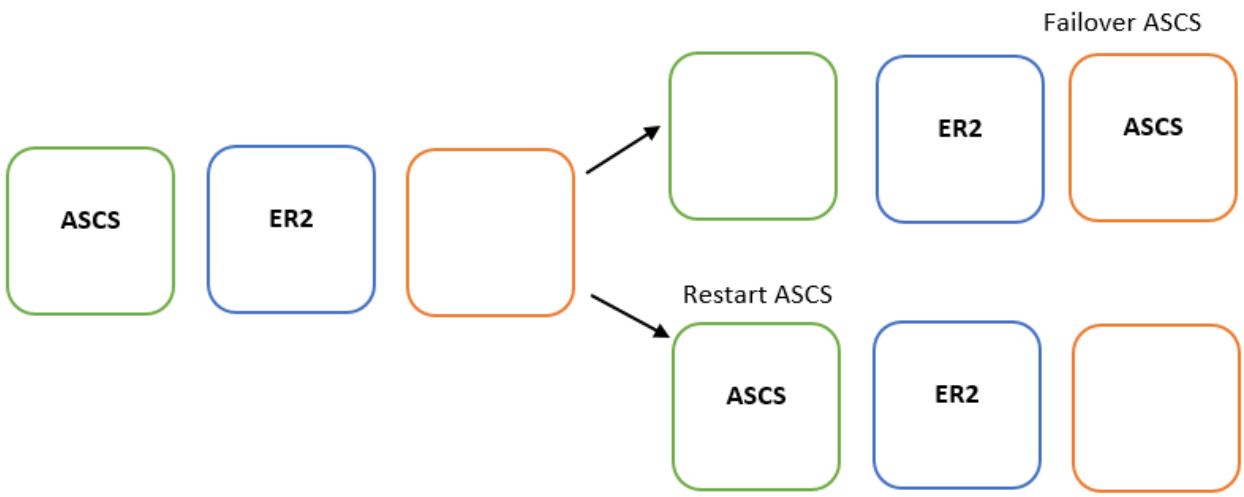
The message server is a Single Point of Failure (SPOF) within the SAP System. Therefore, it is crucial to secure the message server within a switchover group.

Purpose of this test is it, to verify, that in case of a failure of the message server, the HA-Solution detects the failure properly, and triggers a switchover or restart correctly if necessary.

The SAP Startup Framework offers by default a mechanism to restart the message server automatically. Details can be found here.

This test also has the purpose to verify, that these capabilities work well together with the HA-Solution and do not interfere in an unwanted manner.

Illustration (Example with 3 nodes):



Prerequisites:

- This test is only to be executed, in case the message server is not configured to be restarted by the built-in restarting functionality. In order to check that, verify that in the start profile the parameter "Start_Program" is used to start the message server. Otherwise test TEC09 has be executed.
- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated. The HA Setup with 3 or more cluster nodes must be used.

Test procedure:

1.) Kill the message server process on operating system level.

Expected outcome:

Two reactions are allowed for the HA-Solution:

3.) The HA-Solution restarts the ASCS. This means that the ASCS instance remains on the same node as before. Only if the restart fails a certain amount of times, a failover of the ASCS is triggered. The target node must not be the node with the active ER2.

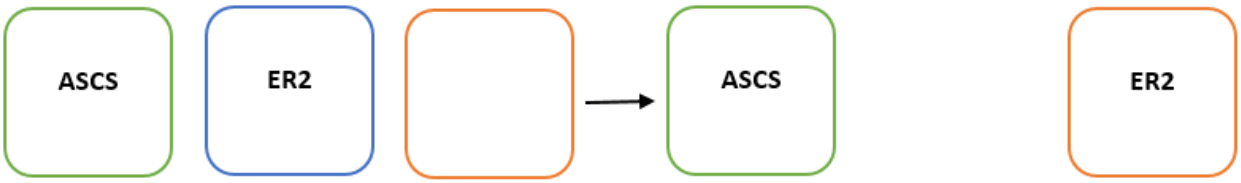
4.) Since the message server is not supposed to be restarted by the SAP Startup Framework, the HA-Solution should trigger a failover of the ASCS instance automatically. The target node must not be the node with the active ER2.

2.3.11 Test TEC11: Enqueue Replicator 2 moves correctly under load in case of hardware or OS failure

Purpose of the test:

The HA-Solution is usually also setup to take care about hardware outages and/or failures of the operating systems. Therefore, this test checks under load whether an unexpected hardware or OS failure triggers a restart of an ER2 instance on a different node.

Illustration (Example with 3 nodes):



Prerequisites:

- Verify that the HA Test Tool is installed, that the SAP system runs in a stable mode, and that HA-Solution is activated. The HA Setup with 3 or more cluster nodes must be used.

Test procedure:

1.) HA-Partner provides Tester with the configuration, event and check files for the test according to the templates of chapter 3.2.

2.) In case the hardware or OS failure is triggered manually and not by the HA Test Tool itself, the event parameters must be commented out in the configuration file of the tool (TEC11Config.properties).

3.) Tester runs the HA Test Tool to trigger a fail-over either by the tool itself or by manually causing a hardware or OS failure during the high load phase of the tool. In the latter case, make sure that the fail-over finishes during the high load phase of the tool.

Expected outcome:

- The HA-Solution automatically triggers a failover of the ER2 to another cluster node, but not using the node with the active ASCS.
- The HA Test Tool terminates with exit code 0.

2.3.12 Test TEC12: Starting and stopping of SAP resources work correctly

Purpose of the test:

Operating the SAP-System within a HA-Setup is crucial. Starting and Stopping an instance is supposed to work correctly, also when the vendor specific shared library or respective scripts are deployed.

Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated.

Test procedure:

1.) Fill the Enqueue-Locktable as described in chapter 3.1.

2.) Use SAP MC/MMC to stop and restart dialog instance 1.

3.) Use SAP MC/MMC to stop and restart dialog instance 2.

4.) Use SAP MC/MMC to stop both dialog instances.

5.) Use SAP MC/MMC to stop the ASCS while all dialog instances are shut down.

6.) Use SAP MC/MMC to start the ASCS.

7.) Use SAP MC/MMC to start both dialog instances.

8.) Use SAP MC/MMC to stop and start the ASCS while both dialog instances are up and running.

Expected outcome:

- The HA-Solution got properly informed thru shared library and corresponding actions were handled correctly.
- The status information within the HA-Solution is updated accordingly.
- Instances can be stopped and restarted via SAP MC/MMC without triggering a failover.
- The Enqueue-Locktable has still the same content than in step 1 (see appendix 3.1: How to check whether the lock table is restored correctly). If not, it's an error of the enqueue mechanism, which should be reported to the enqueue development.

2.3.13 Test TEC13: Outage / Restart of the SAP Start Service is handled correctly

Purpose of the test:

The SAP Start Service is a crucial part of each instance of the SAP System and controls the processes running within the instance. It is crucial that the HA-Solution communicates via the SAP Start Service to these processes, and especially, that the HA-Interface is working properly.

Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated.

Test procedure:

1.) Restart the SAP Start Service of the ASCS manually via SAP MMC/MC. HA-Solution should register the restart of the SAP Start Service; however, no failover of the ASCS is triggered.

2.) Kill SAP Start Service of the ASCS manually on operating system level. The SAP Start Service is restarted automatically.

Expected outcome:

- The HA-Solution should get aware of the fact, that the SAP Start Service was killed and restarted, but does NOT trigger a failover of the ASCS.
- Alternatively, the HA-Solution can trigger a failover of the ASCS instance. In this case, the HA-Solution must ensure that all remaining processes and resources are stopped correctly on the respective host before switching over to the new cluster node. Also, the enqueue lock table must not be lost.
  However, restarting the sapstartsrv is the preferred way to handle the situation.

Comments:

- Please view the logfile in the HA vendor product for these actions.

2.3.14 Test TEC14: Automated ASCS restart in place works correctly

Purpose of the test:

Operating the SAP-System within a HA-Setup is crucial. Starting and Stopping an instance is supposed to work correctly, also when the vendor specific shared library or respective scripts are deployed. Especially, for the Remote Kernel Switch (RKS) is it essential that stopping and starting the ASCS occurs in place without having a significant delay between the stop and the start command.

Prerequisites:

- Verify that the SAP system runs in a stable mode, and that the HA-Solution is activated.

Test procedure:

1.) HA-Partner provides Tester with a script for the respective operating system that does the following:
```
do 5 times
   sapcontrol [-prot protocol] [user/psw] -host <ASCS host> -nr <ASCS systemnumber>
              -function StopWait 120 1
   sapcontrol [-prot protocol] [user/psw] -host <ASCS host> -nr <ASCS systemnumber>
              -function StartWait 120 1
enddo
```

2.) Tester runs the script.

Expected outcome:

- No errors occur during the run of the script.
- The ASCS is stopped and started 5 times on the same node. No failover is triggered.

2.3.15 Test TEC15: Execution of the Rolling Kernel Switch is possible

Purpose of the test:

The Rolling Kernel Switch is designed to update compatible Kernel Patches without any system downtime. This is in particular important for customers, who invested into a recommended HA-Setup for their SAP Systems setup.

The purpose of this test is to ensure that the rolling kernel switch procedure can be performed in the HA-

Setup without any system downtime.

Further information about the rolling Kernel Switch can be found in SAP notes 953653 and 2077934.

Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated. The HA vendor software should not control the ABAP-Dialog instances.
- The installed SAP kernel should have a compatible patch level before the latest available in order to be able to run this test.
- Make yourself familiar with the rolling Kernel Switch procedure as described in SAP note 953653.

Test procedure:

1.) Apply newer kernel version in DIR_CT_RUN as actually running in DIR_EXECUTABLE.

2.) If there is an SAP note for the HA-Solution that is appended to SAP note 2077934, and this note describes additional actions as prerequisites, perform these actions. Not allowed is to

   a. Switch off the cluster

   b. Skip step 3.)

3.) Tester runs the Rolling Kernel Switch via the command
   sapcontrol [-prot <protocol>] [-user <sidadm psw>] -host <RKS_host> -nr <RKS_INSTANCE_NR> -function UpdateSystem 120 300 1

4.) If there is an SAP note for the HA-Solution as described in step 2.), and there are actions described to be done after the automated RKS switch, perform these actions.

Expected outcome:

- All instances run on a new kernel patch level without the need for complete system downtime.
- The HA-Solution should get aware of the fact that the clustered instances of the SAP system were stopped and started but does NOT trigger a failover of these instances.

Comments:

- For the MS Windows platform, following profile parameter setting is not allowed: service/norestart=1.

2.3.16 Test TEC16: Simulating an upgrade works correctly using the Maintenance Mode (if the functionality is available)

Purpose of the test:

Whenever the SAP system needs to be updated, it must be ensured, that the HA-Solution does not interfere with the deployment tools of SAP. If the Maintenance Mode is used, it must be assured that the HA-solution is temporarily disabled by Maintenance Mode calls while the SAP-System is updated.

Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated.
- Provide a HA-solution specific SAP note for the Maintenance Mode. The SAP note should contain

   a. a reference to the general SAP note 2464065 for the Maintenance Mode.

b. the minimum version of the HA Solution that is required for the Maintenance Mode.

c. information about error handling in case of upgrade problems when the Maintenance Mode is switched off (only if more actions than described in SAP note 2464065 are necessary).

d. a description how to check whether the Maintenance Mode is (de)activated on the system.

- Adapt the scripts sumSequence.bat resp. sumSequence.sh according to your environment. The scripts are contained in the file MaintMode v1.01.zip.

Test procedure:

1.) Tester to check that there is a HA-solution specific SAP note for the Maintenance Mode with the content specified above in section 'Prerequisites' is available.

2.) HA-Partner to provide the adapted script (sumSequence.bat or sumSequence.sh)

3.) Tester to execute the script

4.) Tester to execute test TEC05: Irrecoverable outage of the Enqueue Server 2 is handled correctly. Different to the description of TEC05, it is allowed to use the HA Setup with 3 or more cluster nodes.

Expected outcome:

- After step 3): The outcome of the single steps of the script sumSequence.bat/sh are consistent with the expected outcome described in the script.
- After step 4): If a HA setup with 2 cluster nodes is used, the outcome is as expected by test TEC05. If the HA Setup with 3 or more cluster nodes is used, the expected outcome must be as described in TEC08: ASCS moves correctly in case of hardware or OS failure.

2.3.17 Test TEC17: Simulating an upgrade works correctly (if no Maintenance Mode is available)

Purpose of the test:

Whenever the SAP system needs to be updated, it must be ensured, that the HA-Solution does not interfere with the deployment tools of SAP. If no Maintenance Mode is used, it must be described, which actions must be performed to temporarily disable the HA-solution while the SAP-System is updated.

Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated.

Test procedure:

1.) HA-Partner to provide instructions to temporarily disable the HA-Solution to prevent unwanted failovers along a software maintenance activity.

2.) Tester to execute the necessary actions

3.) Tester to stop the complete system manually via SAP MC/ SAP MMC; additionally, stop the SAP Start Service via sapcontrol (function StopService)

4.) After few minutes tester to start the SAP Start Service via sapcontrol (function StartService); start the complete system manually via SAP MC/ SAP MMC

5.) HA-Partner to provide instructions to enable the HA-Solution after a software maintenance activity.

6.) Tester to execute the necessary actions

7.) Tester to execute test TEC05: Irrecoverable outage of the Enqueue Server 2 is handled correctly.

Different to the description of TEC05, it is allowed to use the HA Setup with 3 or more cluster nodes.

Expected outcome:

- After step 3): The SAP System stops, and neither a restart nor failover of the ASCS is triggered.
- After step 7): If a HA setup with 2 cluster nodes is used, the outcome is as expected by test TEC05. If the HA Setup with 3 or more cluster nodes is used, the expected outcome must be as described in TEC08: ASCS moves correctly in case of hardware or OS failure.

Comments:

If the HA-Solution does not provide a respective functionality to temporarily disable and enable the HA-Solution, it is ok to provide a document which describes on how to temporarily disable and enable HA-Solution in case of SW maintenance activities.

If it is not possible or very difficult to temporarily disable and enable the HA-Solution, documentation must be provided, how to perform any software maintenance activities in the fail-over environment.

Any document needs to be accessible for customers and must be linked in the respective SAP support note for the HA-Solution.

All SAP related software maintenance activities that are not possible at all must be documented as well.

2.3.18 Test TEC18: Simulating cluster failure works correctly (if the functionality is available)

Purpose of the test:

Certain HA-Solutions offer the functionality to simulate a failure of a cluster resource. If the to be certified HA-Solution offers such a functionality, the functionality has to work properly also for any SAP related cluster resources. This test verifies the correctness of this functionality, if available.

Prerequisites:

- Verify that the SAP system runs in a stable mode, and that HA-Solution is activated.

Test procedure:

1.) For all SAP related cluster resources, the tester simulates the failure of the resource

Expected outcome:

- The simulation results in the expected action (e.g. automatic restart, or fail-over of the ASCS)

3 Appendix

3.1 How to check whether the lock table is restored correctly

In order to test whether a lock table is restored correctly the following procedure can be used:

1. Read the ENQ-Table by using the command `enq_admin` [Optional step]
2. Create several ENQ-Locks by using the command `enq_admin`
3. Re-read the ENQ-Locks by using the command `enq_admin` before the fail-over
4. Trigger the fail-over of the ASCS-Instance
5. Re-read the ENQ-Locks by using the command `enq_admin` after the fail-over
6. Compare the ENQ-Table before and after the fail-over
7. Double-check the locks held by the Enqueue Replicator 2 (ER2)

Remark: `enq_admin` has to be called on one of the cluster nodes.

The different steps are described in detail below.

Step 1: Read the ENQ-Table by using the command `enq_admin` [Optional step]

Launch the following command via the command line:

```
enq_admin --locks=*:*:*:*:* pf=<ASCS Instance profile>
```

As result, the current ENQ-lock table is displayed.

Linux / Unix: Depending on the shell you must prevent that '*:*:*:*:*' is resolved by the shell. For example, when working with csh use the command

```
enq_admin --locks="*:*:*:*:*" pf=<ASCS Instance profile>
```

Example of a call to read the locks:

```
enq_admin --locks=*:*:*:*:* pf=C:\usr\sap\SID\SYS\profile\SID_ASCS00_ascs
Enqueue Server 2
LID=(5628/1537938366); RN=(3025494); Current=0; Peak=1004; Maximum=250000

Locks: none

2018-10-25 13:24:43; OK; 'Lock List'; Response=238 usec
==========================================================================================
```

Please note that the ASCS instance profile is to be specified to read the ENQ-table.

Instead of using the command `enq_admin`, you can use the command `sapcontrol EnqGetLockTable` to read the ENQ-Lock table.

Step 2: Create several ENQ-Locks by using the command `enq_admin`

Launch the following command via the command line:

```
enq_admin --set_locks=<number of locks>:X:DIAG::TAB:%u pf=<ASCS Instance
profile>
```

As result, the number of locks to be created, will be created.

For example:

```
enq_admin --set_locks=10:X:DIAG::TAB:%u pf=C:\usr\sap\SID\SYS\profile\SID_ASCS00_ascs
```

```
Enqueue Server 2

2018-10-25 13:49:37; OK; 'Set Locks'; Response=4744 usec
==========================================================================================
```

Step 3: Re-read the ENQ-Locks by using the command `enq_admin` before the fail-over

Re-read the ENQ-lock table as described in Step 1.

```
enq_admin --locks=*:*:*:*:* pf=C:\usr\sap\SID\SYS\profile\SID_ASCS00_ascs
Enqueue Server 2
LID=(5628/1537938366); RN=(3027657); Current=10; Peak=1004; Maximum=250000

Locks:
------------------------------------------------------------------------------------------
-------------------------------------------------------------
| Timestamp                  | Type      | Count 1 | Count 2 | BCK   | Object | User      |
Client | TCODE | Session  | Owner 1 | Owner 2 | Name | Argument |
------------------------------------------------------------------------------------------
-------------------------------------------------------------
| 2018-10-25 13:49:37,730353 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 9        |
| 2018-10-25 13:49:37,729926 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 8        |
| 2018-10-25 13:49:37,729448 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 7        |
| 2018-10-25 13:49:37,729027 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 6        |
| 2018-10-25 13:49:37,728572 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 5        |
| 2018-10-25 13:49:37,728138 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 4        |
| 2018-10-25 13:49:37,727678 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 3        |
| 2018-10-25 13:49:37,727256 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 2        |
| 2018-10-25 13:49:37,726766 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 1        |
| 2018-10-25 13:49:37,726147 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 0        |
------------------------------------------------------------------------------------------
-------------------------------------------------------------

2018-10-25 13:49:52; OK; 'Lock List'; Response=304 usec
==========================================================================================
```

Step 4: Trigger the fail-over of the ASCS-Instance

Trigger an ASCS failover by using your HA-Solution as described in the respective test case.

Step 5: Re-read the ENQ-Locks by using the command `enq_admin` after the fail-over

Re-read the ENQ-lock table as described in Step 1

```
enq_admin --locks=*:*:*:*:* pf=C:\usr\sap\SID\SYS\profile\SID_ASCS00_ascs
Enqueue Server 2
LID=(5628/1537938366); RN=(3028713); Current=10; Peak=1004; Maximum=250000

Locks:
------------------------------------------------------------------------------------------
-------------------------------------------------------------
| Timestamp                  | Type      | Count 1 | Count 2 | BCK   | Object | User      |
Client | TCODE | Session  | Owner 1 | Owner 2 | Name | Argument |
------------------------------------------------------------------------------------------
-------------------------------------------------------------
```

```
| 2018-10-25 13:49:37,730353 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 9        |
| 2018-10-25 13:49:37,729926 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 8        |
| 2018-10-25 13:49:37,729448 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 7        |
| 2018-10-25 13:49:37,729027 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 6        |
| 2018-10-25 13:49:37,728572 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 5        |
| 2018-10-25 13:49:37,728138 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 4        |
| 2018-10-25 13:49:37,727678 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 3        |
| 2018-10-25 13:49:37,727256 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 2        |
| 2018-10-25 13:49:37,726766 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 1        |
| 2018-10-25 13:49:37,726147 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 0        |
-------------------------------------------------------------------------------------------------------
-------------------------------------------------------------

2018-10-25 14:02:25; OK; 'Lock List'; Response=298 usec
=========================================================================================================
```

Step 6: Compare the ENQ-Tables before and after the fail-over

Manually check whether the lock table before the fail-over (result of step 3) contains the same locks than after the fail-over (result of step 5) of the ASCS instance.

Step 7: Double-check the locks held by the Enqueue Replicator 2 (ER2)

In case the ER2 instance was moved to a different node, launch the following command via the command line to read the locks of the replication table:

```
enq_admin --locks=*:*:*:*:* --replicator pf=<ER2 Instance profile>
```

Take care of the warning concerning '*:*:*:*:*' described in step 1.

As result, the current ER2-lock table is displayed.

For example:

```
enq_admin --locks=*:*:*:*:* --replicator pf=C:\usr\sap\SID\SYS\profile\SID_ERS10_ers
Enqueue Replicator 2
LID=(5628/1537938366); RN=(3030442); Current=10; Peak=1004; Maximum=250000

Locks:
-------------------------------------------------------------------------------------------------------
-------------------------------------------------------------
| Timestamp                  | Type      | Count 1 | Count 2 | BCK   | Object | User      |
Client | TCODE | Session  | Owner 1 | Owner 2 | Name | Argument |
-------------------------------------------------------------------------------------------------------
-------------------------------------------------------------
| 2018-10-25 13:49:37,721281 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 9        |
| 2018-10-25 13:49:37,720856 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 8        |
| 2018-10-25 13:49:37,720369 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 7        |
| 2018-10-25 13:49:37,719935 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 6        |
| 2018-10-25 13:49:37,719495 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 5        |
| 2018-10-25 13:49:37,719059 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 4        |
| 2018-10-25 13:49:37,718608 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |          | TAB  | 3        |
```

```
| 2018-10-25 13:49:37,718172 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 2        |
| 2018-10-25 13:49:37,717707 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 1        |
| 2018-10-25 13:49:37,717185 | Exclusive |       1 |       0 | false | E_FILL | FILL_USER | 000
| SFILL | Tx_Uy_Mz | DIAG    |         | TAB  | 0        |
-------------------------------------------------------------------------------------------------
------------------------------------------------------------

2018-10-25 14:29:38; OK; 'Lock List'; Response=263 usec
=================================================================================================
```

Manually check whether the replication table contains the same locks than the lock table of the ASCS instance (result of step 3 or 5).

3.2 Configuration, Event and Check Files for HA Test Tool

The HA Test Tool and the configuration, event and check files that are needed for the tests with the SAP HA Test Tool are provided by the SAP ICC.

The standard HA Test Tool package is named HATool_vxyz.zip (xyz is the version number of the tool). Unpacking the file you find the documentation that describes the usage of the tool in directory HATool/doc.

For the tests needed for the certification unpack the file HAToolContent_CertiS4_1.0.zip to the same directory on which the client of the HA Test Tool is installed. Following files are delivered:

…/HATool	Base directory of the HA Test Tool at client side
|	
+ …../check	Contains the files to check the results of test runs:
|	TECxxCheck.xml
+ …../config	Contains the property files for the tool:
|	TECxxConfig.properties, TECxxEvent.properties, Login.properties
+ …../event	Contains script file templates to trigger HA events
	TECxxEvent.bat, TECxxEvent.sh

Take care, that sapcontrol is installed as described in section 2.1.3 (Installing the HA Test Tool). Adapt Login.properties, TECxxEvent.properties and TECxxEvent.bat/sh to your environment. TECxxCheck.xml and TECxxConfig.properties must not be changed unless it is explicitly allowed in the test description.